AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM359
Systems Penetration Testing

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: (including this cover sheet)	(3)	Time Allowed: 2 h

Instructions:

- Total Marks: 60
- Total Number of Questions:6
- Marks Distribution:
  - Question 1: 10 marks
  - Question 2: 10 marks
  - Question 3: 10 marks
  - Question 4: 10 marks
  - Question 5: 10 marks
  - Question 6: 10 marks
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.

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Question 1 [10 marks]

For each question fill in the blank space with an appropriate word, acronym, name or phrase. For each blank space you must give only one answer. However, there may be more than one correct answer. Each question is worth 1 mark.

1. ---------- is the process of taking organized and carful steps when reacting to a security incident or cyberattack.
2. ----------is pretending to be a legitimate or authorized person and using the phone or other communication medium to mislead targets and trick them into revealing information.
3. ---------is dynamically allocated at runtime during the execution of the program and it stores program data.
4. Attackers can identify the OS running on the target machine by looking at the ------- -------- and --------- ----- in the IP header of the first packet in a TCP session.
5. --------- is the process of enumerating user accounts and devices on a target system.
6. Attackers use google ---- ---------- in google advanced search operators to extract sensitive information about their target.
7. SNMP read community string is a ---------- by default, it does not allow for the viewing of the device/system configuration
8. ---------- is a major security problem with Telnet.
9. ---------- is the practice of hiding a secret message inside of (or even on top of) something that is not secret.
10. ---------- port is used by the NetBIOS Name service (NBNS).

Question 2 [10 marks]

1. Answer the following questions based on the screenshot given below,

   A. What is an attack type? [2 mark]
   B. What is the main objective of the command an attacker used? [2 mark]
   C. What is sensitive information exfiltrated from this attack? [2 mark]
   D. What is the next step the attacker may take? [2 mark]
   E. How to mitigate this attack type? [2 mark]

```
                    telnet 209.165.200.224 80
Trying 209.165.200.224...
Connected to 209.165.200.224.
Escape character is '^]'.
<type anything to force an HTTP error response>
HTTP/1.1 400 Bad Request
Server: nginx/1.12.0
Date: Wed, 17 May 2017 14:27:30 GMT
Content-Type: text/html
Content-Length: 173
Connection: close
<html>
<head><title>400 Bad Request</title></head>
<body bgcolor="white">
<center><h1>400 Bad Request</h1></center>
<hr><center>nginx/1.12.0</center>
</body>
</html >
Connection closed by foreign host.
analyst@secOps ~]$
```

Question 3 [10 marks]

State whether the statements are true or false and correct the wrong statement

1. ISO/IEC 27001:2013 is a proprietary information security standard for organizations that handle cardholder information for major debit, credit, and prepaid cards.
2. Non-repudiation guarantees that the sender of a message cannot later deny having sent the message and that the recipient cannot deny having received the message.

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3. One of IP Spoofing Countermeasures, encrypt all the network traffic using cryptographic network protocols such as Telnet and HTTP.
4. SNMP read community string is a private by default, it does not allow for the viewing of the device/system configuration.
5. Information Assurance refers to the assurance that the CIA and authenticity of information systems is protected during the usage, processing, storage of information

Question 4 [10 marks]

1. What is Enumeration, list four techniques for enumeration? (5 Marks)
2. Explain in detail by using five different methods, how to evade IDS/firewalls? [5 Marks]

Question 5 [10 marks]

1. ----------is a security strategy in which several protection layers are placed throughout an information system
   a. Defense in depth
   b. Least privilege
   c. Zero trust
   d. Dumpster Diving
2. ----------is the art of discovering and rendering covert messages using steganography.
   a. Steganography
   b. Confidentiality
   c. Steganalysis
   d. Integrity
3. ----------is an open-source intelligence gathering framework that is focused on gathering information from free tools or resources
   a. Defense in depth framework
   b. Google framework
   c. OSINT framework
   d. Nmap framework
4. ----------in which attackers recover passwords from hashes with specific set of characters based on some information known to the attacker
   a. Mask attack
   b. Recover attack
   c. Hashing attack
   d. Password attack
5. Obtaining an accurate service version number allows attackers to----------
   a. determine the vulnerability of the target system to exploit
   b. determine the service version to update
   c. determine the service version to upgrade
   d. determine the application version and save it

Question 6 [10 marks]

Explain in detail the following: (10 marks)

1- What is Mask attack? [2.5 marks]
2- What are common reasons behind the existence of vulnerabilities? [2.5 marks]
3- What is Service version discovery? [5 marks]

End of Questions

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